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A pilot study for whole proteome tagging in *C. elegans*

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eLife Assessment

The nematode *C. elegans* is an ideal model in which to achieve the ambitious goal of a genome-wide atlas of protein expression and localization. In this paper, the authors explore the utility of a new and efficient method for labeling proteins with fluorescent tags, evaluating its potential to be the basis for a larger, genome-wide effort that is likely to be very **useful** for the community. While the evidence for the method itself is **solid**, carrying out this project at a large scale will require significant additional feasibility studies.

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Abstract

Tagging all proteins encoded by an animal genome with a fluorescent tag would open many windows to the discovery of unexpected patterns of protein expression and localization. To scale such an approach, it would be beneficial to introduce multiple, spectrally distinct fluorophore tags in parallel. As a first step in this direction, we undertook a pilot study in the nematode *C. elegans*, in which we set out to tag 30 different genetic loci with three different fluorophores, with 3 tags being introduced at a time. By choosing essential genes, predicted based on transcriptomics to cover a range of expression levels, we explore issues relating to disrupting gene function and visibility of tagged proteins. We demonstrate that such a tagging approach is highly efficient and indeed reveals unanticipated patterns of cellular sites of expression, as well as subcellular protein localization. We hope that this pilot study will motivate attempts to scale this tagging approach to more loci and, ultimately, the whole genome.

Introduction

Gene transcript-based expression atlases have provided invaluable insights into animal biology. The next frontier lies in generating whole animal protein expression atlases with single cell resolution that capture the cellular and subcellular specificity of protein expression and localization and visualize the dynamics of these parameters during development and during changes in internal and external states. The establishment of protein expression atlases is particularly urgent in light of the generally appreciated discordance between transcript detection – often used to infer cellular functions – and protein expression, a reflection of multifarious layers of posttranscriptional gene regulation (Liu et al., 2016 [↗](#); Schwanhäusser et al., 2011 [↗](#); Vogel and Marcotte, 2012 [↗](#)).

One means to acquire protein expression and localization on a single cell level with whole proteome coverage lies in tagging all genes in the genome with a fluorescent reporter. This approach has been pioneered in yeast (Huh et al., 2003 [↗](#)), where chromosomally GFP-tagged strains provided the first cellular map covering ~75% of the yeast proteome, assigning localization for 70% of previously unlocalized proteins while uncovering the compartmental logic underlying protein function and interactions. Tagging proteins in multicellular organisms provides a host of

new insights since the differentiation of cells into distinct types and their assembly into tissues deploys many more proteins and encompasses the establishment and dynamic control of more complex protein expression and localization patterns.

We envision the tagging of the entire proteome of a multicellular organism to be most feasible in the nematode *C. elegans*. CRISPR/Cas9 genome engineering is now routinely used to tag individual genetic loci in *C. elegans* with a fluorophore. So far, around 1,500 genes have been tagged already by individual labs (Leyhr et al., 2025 [\[1\]](#)). With recent procedural advances (Eroglu et al., 2023 [\[2\]](#); Ghanta and Mello, 2020 [\[3\]](#); Paix et al., 2015 [\[4\]](#)), such engineering is highly effective and rapid, given the short generation time of *C. elegans*. The transparency of the organism facilitates imaging of fluorophores in live animals. Advances in light-sheet fluorescence microscopy, such as dual-view inverted selective plane illumination microscopy (diSPIM), have enabled high-resolution, 3-dimensional live imaging of dynamic fluorophore localization patterns throughout all stages of development (Kumar et al., 2014 [\[5\]](#); Wu et al., 2017 [\[6\]](#)). Super-resolution imaging approaches such as image scanning microscopy (ISM) (Huff, 2015 [\[7\]](#)) and stimulated emission depletion (STED) microscopy (Rankin et al., 2011 [\[8\]](#)) enable increasingly finer localization of proteins in live worms. Further resolution can also be achieved in fixed worms with expansion microscopy, without the requirement for specialized microscopy equipment (Yu et al., 2022 [\[9\]](#)). Extensive anatomical atlases are either already available (Altun et al., 2002 [\[10\]](#)) or in the process of being generated (Cook et al., 2019 [\[11\]](#); Santella et al., 2022 [\[12\]](#); Witvliet et al., 2021 [\[13\]](#)), allowing the mapping of fluorophore localization patterns onto anatomical features at electron microscopical scales (Vergara et al., 2021 [\[14\]](#)).

There is also ample precedent in the *C. elegans* literature that reporter tagging results in unanticipated insights. To cite two recent examples from our own lab, we found that Golgi-resident proteins displayed cell type-specific expression patterns not anticipated by transcriptomic analysis (Leyva-Díaz et al., 2025 [\[15\]](#)). As another example, the subcellular localization pattern of a DEG/ENaC channel in enteric neurons indicated unexpected functional insights into the protein (Bayer et al., 2025 [\[16\]](#)). It is also notable that the widely appreciated discordance of mRNA and protein expression levels is evident in *C. elegans* (Grün et al., 2014 [\[17\]](#); Reilly et al., 2022 [\[18\]](#), 2020 [\[19\]](#)), attesting to the need to extend currently existing transcriptome atlases (Ghaddar et al., 2023 [\[20\]](#); Packer et al., 2019 [\[21\]](#); Roux et al., 2023 [\[22\]](#); Taylor et al., 2025 [\[23\]](#), 2021 [\[24\]](#)) with protein expression atlases.

As a proof of concept for scaling such a CRISPR/Cas9-engineered project to the whole genome, we report here the results of a pilot study in which we aimed to CRISPR/Cas9-engineer reporter tags into 30 genomic loci, generated through simultaneous tagging of three genes at a time. A subset of these 30 genes reveals unexpected patterns of expression and localization, illustrating how protein tagging approaches provide a route for new discoveries. We discuss here several aspects of such tagging approaches, highlighting promises, as well as limitations.

Results

Pipeline for pooled high throughput tagging by CRISPR/Cas9

Recent methods for CRISPR/Cas9 mediated genomic insertions have pushed efficiencies to sufficient levels to simultaneously insert multiple fluorophores (mNeonGreen and mScarlet) as well as a co-CRISPR marker (*dpy-10*) at three independent loci in a single injection (Eroglu et al., 2023 [\[2\]](#)). Thus, in principle current methods could allow tagging of at least 3 independent loci in one injection, particularly if a co-CRISPR marker is omitted. Endogenously tagged genes can be detected by fluorescence stereomicroscopy if the fluorophores are expressed at sufficiently high levels and hence are readily visible as heterozygotes in F1 by visual screening without co-CRISPR markers. We reasoned that ubiquitously and highly expressed genes could be used as insertion markers and pooled with low or modestly expressed genes, with a first round of visual screening of the highly expressed genes followed by targeted PCR or higher power imaging of the lower expressed genes (Fig. 1A [\[25\]](#)).

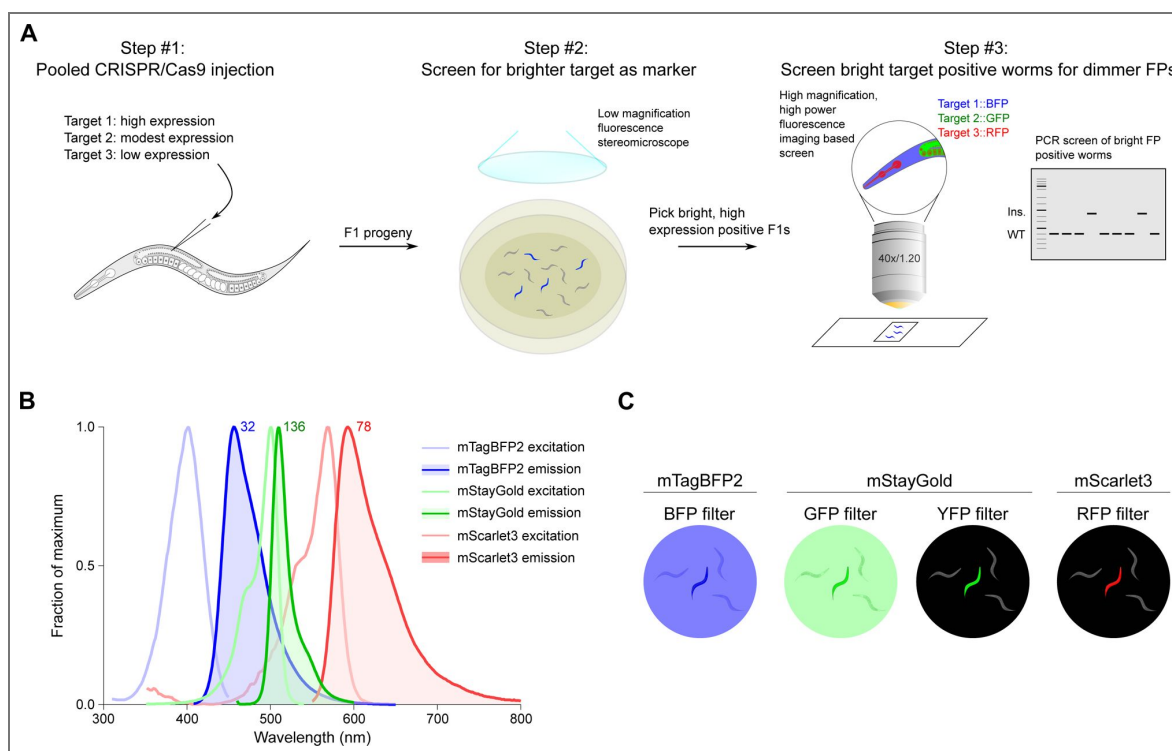


Figure 1. Pipeline for triple pooled CRISPR-Cas9 mediated gene tagging.

(A) Schematic of injection and screening strategies. FP, fluorescent protein. **(B)** Excitation and emission spectra of selected fluorescent proteins used in the pilot study. Theoretical brightness of the fluorophores, calculated as a product of the extinction coefficient and quantum yield, is annotated as numbers adjacent to the respective emission spectra. The spectra and brightness values were adapted from FPbase (Lambert, 2019). **(C)** Schematic of the practical visibility of fluorophores observed under a fluorescent stereomicroscope with different filter sets in live worms on NGM plates.

To facilitate screening by fluorescence stereomicroscopy and co-imaging, we chose the three currently brightest fluorescent proteins in the blue, green and red emission spectra: mTagBFP2 (Subach et al., 2011 [DOI](#)), mStayGold(J) (Ando et al., 2024 [DOI](#)) and mScarlet3 (Gadella et al., 2023 [DOI](#)), respectively (Fig. 1B [DOI](#)). These tags are significantly brighter than first generation fluorophores, thereby allowing the detection of even very lowly expressed proteins. Autofluorescence in *C. elegans* is highest in blue and green wavelengths, and lowest in red. Furthermore, we noted that nematode growth medium (NGM) plates seeded with OP50 also contribute to background fluorescence with conventional blue and green filters on a fluorescent stereomicroscope (Fig. 1C [DOI](#)). In contrast, we observed minimal background with yellow and red emission filters from worms as well as bacteria. We also noted from existing tagged strains that mStayGold could be observed brightly with YFP filters, which, combined with the minimal background, yielded a greater contrast with untagged worms enabling easier screening of mStayGold positive worms compared to conventional GFP filters (Fig. 1C [DOI](#)). However, given the suboptimal excitation and emission of visualizing mStayGold with a YFP filter, mScarlet3 in practice yielded the greatest signal-to-noise ratio for visual screening. Thus, to balance fluorescence signal-to-noise ratios with predicted gene expression, we assigned the highest expressed genes to be tagged with mTagBFP2, modestly expressed genes with mStayGold, and lowest expressed genes with mScarlet3.

In the pilot experiment, we sought to establish three major principles: (1) the endogenous expression ranges at which fluorophores could be utilized for visual screening; (2) success rate of triple pooled tagging; (3) functionality of tagged proteins. We thus selected 30 genes across a variety of bulk transcript expression ranges which are predicted to be ubiquitously expressed based on molecular function or single cell RNA sequencing (scRNA-seq) data (Table 1 [DOI](#), Fig. 2A, B [DOI](#)) (Taylor et al., 2021 [DOI](#)). The proteins are generally predicted to be broadly expressed and cover a very wide range of predicted cellular functions and predicted subcellular localization patterns (nuclear, cytosolic, transmembrane). 26 of 30 genes are annotated to display lethal, sterile, or other phenotypes by RNAi, enabling the assessment of tagged protein functionality.

The *C. elegans* genome encodes around 20,000 genes. Approximately 6,700 marker genes would need to be sufficiently highly expressed to be used as visible markers to group with the entire genome in sets of three. In bulk RNA-seq of N2 worms at L4 stage, 6,804 genes are expressed above 20 fragments per kilobase of transcript per million mapped reads (FPKM) (Eroglu et al., 2024 [DOI](#); Gerstein et al., 2010 [DOI](#); Harris et al., 2020 [DOI](#)). Based on our prior experience comparing bulk transcript FPKM with visibility, we divided these genes into three subsets to assess the detection limit of the three fluorophores: ≥ 500 RPKM for mTagBFP2; 100-500 RPKM for mStayGold; and 18-100 RPKM for mScarlet3.

We tagged genes either at their N- or C-terminus, primarily aiming to capture as many predicted isoforms as possible while retaining a potential guide RNA target sequence within 10 nucleotides of the start or stop codon. Where both termini satisfied these conditions, we selected the C-terminus over the N-terminus, as previous studies have suggested that C-terminal tags are less likely to disrupt protein localization and expression (Davidi et al., 2019 [DOI](#); Palmer and Freeman, 2004 [DOI](#)) and that genome wide tagging attempts in yeast have successfully used C-termini (Huh et al., 2003 [DOI](#)) with high concordance to prior annotated data. However, we acknowledge that subclasses of proteins such as peroxisomal proteins, C-tail- or glycosylphosphatidylinositol-anchored proteins, lipidated proteins and proteins with retrieval motifs, may only show correct localization when tagged at the N-terminus (Yofe et al., 2016 [DOI](#)).

Efficient tagging across loci, expression levels and target pools

We co-injected the 30 selected targets as 10 pools (Table 1 [DOI](#)), with each pool being injected into 5 worms. Overall, we successfully isolated 8/10 mTagBFP2 tags, 9/10 mStayGold tags, and 7/10 mScarlet3 tags (Fig. 3A, B [DOI](#)). All but one tag (*cox-6B::mTagBFP2*) was visible in the F1 generation of injected P0 animals, and these were subsequently isolated among F2 worms positive for the other tags in the pool. In 4 pools, we isolated all 3 tags within the same worm, and in an additional experiment all 3 tags but in pairs. The remaining tags were isolated as pairs or singles. Even when

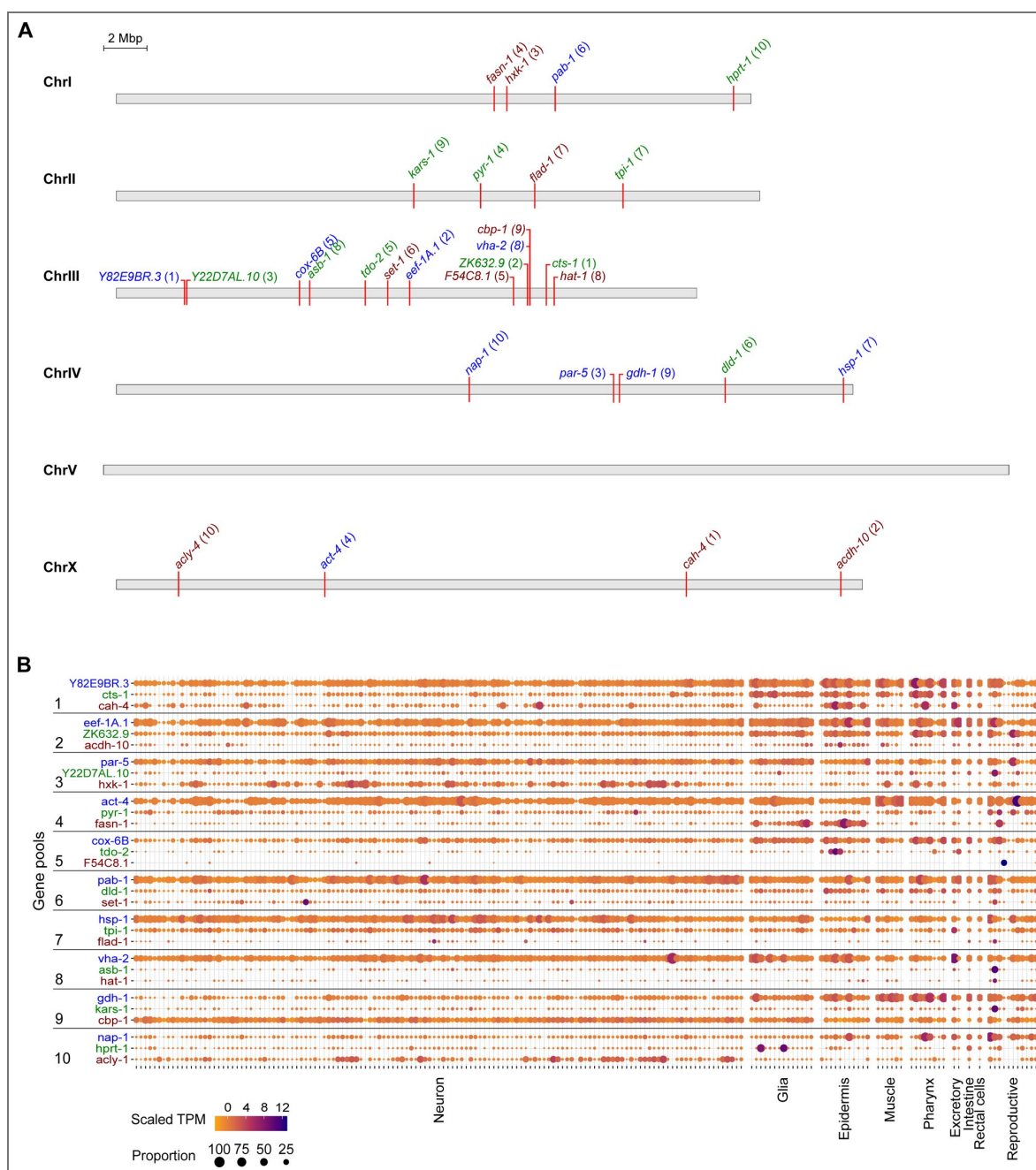


Figure 2. Selection of genes and predicted transcript expressions across tissues.

(A) Location of selected genes on *C. elegans* chromosomes. Numbers in parentheses indicate the injection pool the gene was assigned to. Color of text indicates the fluorophore assigned to the gene: blue, mTagBFP2; green, mStayGold; red, mScarlet3. (B) Transcript expression levels of the selected genes across major tissues of *C. elegans* as measured by single cell RNA sequencing of adult hermaphrodites. Data was adapted from the CeNGEN adult hermaphrodite set (Taylor et al., 2021).

Group	Gene	Chr	Class	Function	WormBase L4 F1 Terminals	Tag	Success?	RNAi Phen	Tag dissecting visible?	Rationale
1	Y82E9B.3	III	ATP synthase membrane subunit c	ATP synthesis & ETC	7258.1 N	m18FP2	No	Let		Highest expressed ETC component
	cts-1	III	citrate synthase	Citrate cycle TCA cycle	465.7 C	m5C3	Yes	Let	Yes	Highest expressed citrate cycle
	cat-4	X	carbonic anhydrase	Nitrogen metabolism	104.8 C	m18FP2	Yes	Let	Yes	Highest expressed nitrogen fixation
	cat-4	III	elongation factor 1 alpha 1	Translation	6059 C	m18FP2	Yes	Let	Yes	Highest expressed translation elongation factor
2	ZK632.9	III	Nuclear protein 1	Nucleus	452.4 N	m5C3	Yes	Stp	Yes	Among highest expressed pred. nuclear proteins, uncharacterized
	acdH-10	X	acyl-CoA dehydrogenase	Metabolism	73.5 C	m5C3	Yes	Stp	Yes	Fatty acid metabolism, Acyl CoA Dehydrogenase
	par-5	IV	tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein	Signaling	2121.8 C	m18FP2	Yes	Let	Yes	Highest expressed 14-3-3 gene
3	Y22D7AL.10	III	heat shock protein family E	Proteostasis	363.5 C	m5C3	Yes	Let	Yes	Uncharacterized, highly expressed heat shock protein
	hxt-1	I	glucokinase	Glycolysis / Gluconeogenesis	46.3 C	m5C3	Yes	Let	Yes	Initial enzyme of glycolysis (glucose -> g6p)
	act-4	X	actin beta	Cytoskeleton	1661.6 C	m18FP2	No	Let		Highest expressed actin
4	pyr-1	II	carbamoyl phosphate synthetase 2, aspartate transcarbamylase, and dihydroorotase	Pyrimidine metabolism	58.9 C	m5C3	Yes	Let	Yes	Initial enzyme of pyrimidine synthesis
	fasn-1	I	fatty acid synthase	Fatty acid biosynthesis	38.1 C	m5C3	No	Let		Converts acetyl CoA to fatty acids
	cox-6b	III	cytochrome c oxidase subunit 6B	ATP synthesis & ETC	1650.1 C	m18FP2	Yes	Let	Dim F1, F2 yes	Highest expressed cytochrome c oxidase
5	tkb-2	III	tryptophan 2,3-dioxygenase	W degradation	201.3 C	m5C3	Yes	Oro	Yes	Tryptophan degradation
	F54C8.1	III	hydroxyacyl-CoA dehydrogenase	Fatty acid elongation	34.2 C	m5C3	Yes	Let	Yes	Uncharacterized
	pab-1	I	poly(A) binding protein	Transcript reg	1017.1 C	m18FP2	Yes	Let	Yes	Highest expressed PAB
6	did-1	IV	dihydropyrimidine dehydrogenase	Glycolysis / Gluconeogenesis	199.7 C	m5C3	Yes	Let	Yes	Mitochondrial acetyl coA synthesis from pyruvate
	sp-1	III	lysine methyltransferase	Histone modification	27.1 C	m5C3	No	Let		Highest expressed histone modifier
	hsp-1	IV	heat shock protein family A	Proteostasis	998.7 C	m18FP2	Yes	Let	Yes	Highest expressed heat shock protein
7	tpi-1	II	triosephosphate isomerase	Glycolysis / Gluconeogenesis	180.1 C	m5C3	Yes	Yes		Glycolysis, cytosolic
	flad-1	II	flavin adenine dinucleotide synthetase	Riboflavin metabolism	23.7 Internal, eve	m5C3	No	Oro		Only FAD synthetase (cofactor for histone demethylation by LSD/KDM family)
	vha-3	III	ATPase H+ transporting V0 subunit c	Proton transport	732.3 C	m18FP2	Yes	Let	Yes	Highest expressed V-ATPase
8	asb-1	III	ATP synthase peripheral stalk-membrane subunit b	Oxidative phosphorylation	157.1 N	m5C3	No	Let		ATP synthase
	hat-1	III	histone acetyltransferase 1	Histone modification	23.3 C	m5C3	Yes	Let	Dim	Histone acetyltransferase
	gpb-1	IV	glutamate dehydrogenase	Glutamate / glutamine metabolism	465.2 C	m18FP2	Yes	Stp	Yes	arg synthesis, (cofactor for histone demethylation by JmjC family)
9	gpb-1	IV	glutamate dehydrogenase	Glutamate / glutamine metabolism	465.2 C	m18FP2	Yes	Stp	Yes	Highest expressed aminocycl RNA synthetase
	cbp-1	III	CREB binding protein	Histone modification	22.7 C	m5C3	Yes	Let	Dim	CREBBP/p300, broad transcriptional activator, ubiquitous
	hsp-1	IV	nucleosome assembly protein	Nucleus	500.6 C	m18FP2	Yes	Let	Yes	Uncharacterized, predicted role in nucleosome assembly
10	hprt-1	I	phosphoribosyl transferase	Purine metabolism	101 C	m5C3	Yes	Let	Yes	Purine salvage pathway, IMP synthesis
	acly-1	X	ATP citrate lyase	Citrate cycle TCA cycle	18 N	m5C3	Yes	Let	Yes	Citrate to acetyl CoA and OAA

Table 1. Genes tagged

one or two tags in the pool failed, the remaining tags in the same pool could be independently isolated. Overall, 24 of 30 tags were successfully isolated, all of which were visible with fluorescence stereomicroscopy.

During our screening, we made several observations which enabled us to assess whether failures were due to technical or biological reasons. Our first attempt with pools 4, 6 and 10 yielded no positive progeny for any of the 3 targets in the respective pools despite the injected worms showing a healthy number of viable offspring (Fig. 3C). We reasoned that, given overall efficiency for individual tags, this was due to technical failures rather than the impossibility of functionally tagging all three genes within the pool. Indeed, upon reconstitution of the injection mixes with additional quality control steps (i.e., ensuring that repair templates were of sufficient purity as assessed by proper A260/230 and A260/280 ratios; see Methods), we successfully obtained 6/9 of the originally failed tags (Fig. 3C). In contrast, failures in obtaining one or two of the tags in the pool were generally interpreted as loss of function tags, and in these cases, we frequently observed fluorescent but dead embryos on the plates. While the combination of high brood size with 0/3 successful tags was a good indicator to reattempt individual pools, overall brood size across successful injections did not correlate with efficiency of insertion (Fig. 3D).

Unanticipated cell type specific enrichment of targets

We assessed the expression patterns of the protein fusions by confocal microscopy in adult worms and compared them to expected tissue type specific abundances predicted by stage-matched scRNA-seq datasets (Ghaddar et al., 2023; Taylor et al., 2021). Based on molecular functions in fundamental biological processes, all genes were expected to be ubiquitously expressed. Likewise, scRNA-seq indicated that transcript abundance was ubiquitous and without strong tissue-specific enrichment with few exceptions.

In contrast, we found notable examples of mismatch between expected and observed expression patterns. The translation elongation factor *EEF-1A.1* showed strong germline enrichment whereas it was predicted to be enriched in various cell types including germ cells, epidermis, intestinal and excretory cells by scRNA seq (Fig. 4). The acyl-CoA dehydrogenase *ACDH-10* was somatically broadly and evenly expressed but strongly depleted in the germline, while we expected enrichment in intestinal and epidermal tissues given its role in fatty acid metabolism (Fig. 4). Strikingly, the glucokinase *HXX-1* showed strong enrichment in the gonadal sheath, whereas we expected ubiquitous expression with potential enrichment in glycolysis upregulated tissues such as muscle and neurons (Fig. 5). The lysyl-tRNA synthetase *KARS-1* showed germline enrichment, while it was expected to be evenly distributed across tissues given its role in translation, but was correctly predicted by scRNA seq to be germline enriched (Fig. 6A-C). The nucleosome assembly protein *NAP-1* was highest expressed in the germline, whereas it was predicted to be highest in the distal tip cell and pharynx (Fig. 7A). In contrast, the phosphoribosyl transferase *HPRT-1* and ATP citrate lyase *ACLY-1* were enriched in subsets of glia and neurons, respectively, consistent with scRNA-seq datasets (Fig. 7A, B). The cytochrome c oxidase subunit 6B *COX-6B* was highest enriched in the distal gonad and intestine (Fig. 8A), while scRNA-seq predicted highest expression in the pharynx. The tryptophan 2,3-dioxygenase *TDO-2* showed highly enriched or specific expression in epidermis, whereas its expression predicted by scRNA seq was ubiquitous with modest epidermal enrichment (Fig. 8A, B). The hydroxyacyl-CoA dehydrogenase *F54C8.1*, which we independently isolated from the same pool as *COX-6B* and *TDO-2*, showed high enrichment in sperm (Fig. 8B, C) whereas it was predicted to be enriched in the intestine based on function but correctly suggested by scRNA-seq as highly enriched in or specific to sperm. We note that the gut granule signal in *F54C8.1::mScarlet3* tagged worms was comparable to that observed in untagged N2 worms, indicating that it is background autofluorescence (Fig. 8D). The heat shock protein *HSP-1* was enriched in the germline while predicted to be ubiquitous and evenly expressed by scRNA-seq (Fig. 9). Similarly, the triosephosphate isomerase *TPI-1* was observed to be enriched in neurons, which was not predicted from scRNA-seq data (Fig. 9).

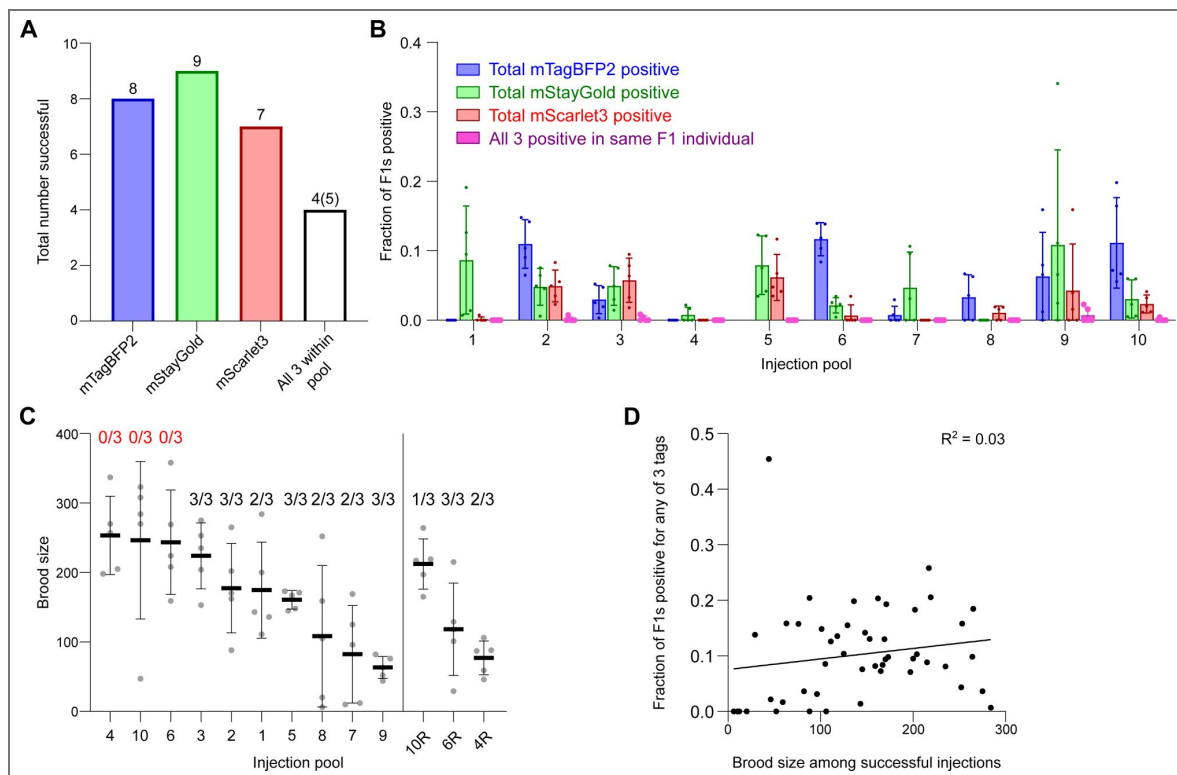


Figure 3. Successful tagging across loci, fluorophores and gene pools.

(A) Total number of targets successfully tagged. White bar denotes all 3 genes within the pool were successfully tagged in any combination. In four pools, all 3 targets were isolated in a single, triple-tagged individual worm. In one pool, we obtained all 3 targets but in doubles. **(B)** Total insertion efficiencies for the indicated targets across injection pools. Dots, fraction of F1s positive for indicated fluorophore per injected P0. Five worms were injected per pool except for pools 4, 10 and 6 which were repeated after initial failures as shown in panel C. Data shown in panel B only represents the repeat runs and omits the original failed injections. Bars, mean $\pm$ SD. **(C)** Viable brood size of injected P0 worms represented as individual points. Lines, mean $\pm$ SD. Numbers above plots, number of successfully isolated tags out of 3 within the pool. 4R, 6R and 10R are re-injections of groups 4, 6 and 10. **(D)** Correlation of viable brood of P0s with fraction of F1 worms positive for any fluorophore tags. Goodness of fit is represented as an R^2 value from a simple linear regression. Slope was not significantly greater than zero ($P = 0.2497$, F-test on Prism).

Overall, numerous genes with ubiquitous and evenly distributed transcript expression showed tissue-specific enrichment not predicted by transcript levels with a general trend towards underestimation of relative protein levels in the germline.

Unanticipated subcellular localization patterns of proteins

Given that our tags were designed to be translationally fused to the proteins of interest, we assessed their subcellular localizations and compared them to localizations annotated experimentally or predicted by homology (Harris et al., 2020 [↗](#)). Most notably, we found two proteins, GDH-1 and KARS-1, detected in mitochondria by mass spectrometry (Guo et al., 2025 [↗](#); Li et al., 2009 [↗](#)), and co-tagged in the same worm, to display mitochondrial-like localizations; however, they did not overlap (Fig. 6B, C [↗](#)). This finding suggests that subsets of mitochondria in the gonad may have varying markers and metabolic specializations. Indeed, a study published around the time of our pilot experiment described metabolically distinct subpopulations of mitochondria depending on cellular energy demand (Ryu et al., 2024 [↗](#)).

Furthermore, we found that the V-ATPase VHA-2, involved in proton transport and predicted to be localized to cell and vesicular membranes and detected in membrane fractions by mass spectrometry (Mawuenyega et al., 2003 [↗](#)), displayed perinuclear localization in the gonad specifically (Fig. 10 [↗](#)). This indicates that cell type-specific nuclear envelopes or endoplasmic reticula may have unique demands for pH control which are yet unexplored, to our knowledge.

In general, subcellular localizations were predicted more accurately than cell type-specific abundance, though we observed additional dynamic localization patterns unanticipated by molecular function. For instance, the DNA-binding nuclear phosphoprotein p8 homolog ZK632.9 was predicted to be nuclear but observed diffusely with modest nuclear enrichment (Fig. 4 [↗](#)). We note that this may be due to its small size of 62 amino acids, which could be influenced by the larger 226 amino acid mStayGold tag. However, the tagged worm does not show any annotated ZK632.9 depletion phenotypes such as sterility (Simmer et al., 2003 [↗](#)), indicating that it is to some extent functional. Additionally, the histone acetyltransferase 1 HAT-1 showed broadly nuclear localization as predicted but was cytoplasmic in early embryos (Fig. 10 [↗](#)), indicating dynamic subcellular localization. Finally, the tryptophan 2,3-dioxygenase TDO-2 with previously unknown localization appeared cytoplasmic (Fig. 8A, B [↗](#)).

Discussion

We have provided proof of concept for the ease and efficiency of tagging at least three proteins at once during a single round of injection. We also find that modern-day fluorophores allow the visualization of proteins at a wide range of expression levels, even at the lowest expression levels tested. Extrapolating our results to annotated transcript levels of *C. elegans* genes, we predict that at least 324 proteins could be directly visualized by mTagBFP2, 1578 with mStayGold, and 7249 with mScarlet3 for high throughput screening using low powered fluorescence stereomicroscopy. At minimum, this could enable annotation of one third of *C. elegans* proteins and at best provide sufficient markers to co-tag the entire genome in groups of 3 genes. In the latter scenario, brighter genes would be used as coCRISPR markers grouped with dimmer genes which would be isolated among bright gene positive worms through alternative screening methods such as PCR or higher magnification, longer exposure fluorescence microscopy (Fig. 1A [↗](#)). Given many genes are not broadly expressed, their bulk transcript levels in whole worms are likely to underestimate expression within subsets of cells, which may aid in their detectability and isolation. Similarly, nuclear localized proteins will likely be easier to detect even at low expression levels, given the concentration of signal in small subcellular compartments. Even with the relatively small number of targets tested, we have observed numerous unanticipated patterns of tissue-specific enrichment and subcellular localization.

By tagging essential genes and in most cases observing no overt effect on viability of the animal, we have also shown that many N- or C-terminal tags do not disrupt protein function in an obvious way. Given most genes we targeted were essential, we could infer the effect of tagging through the

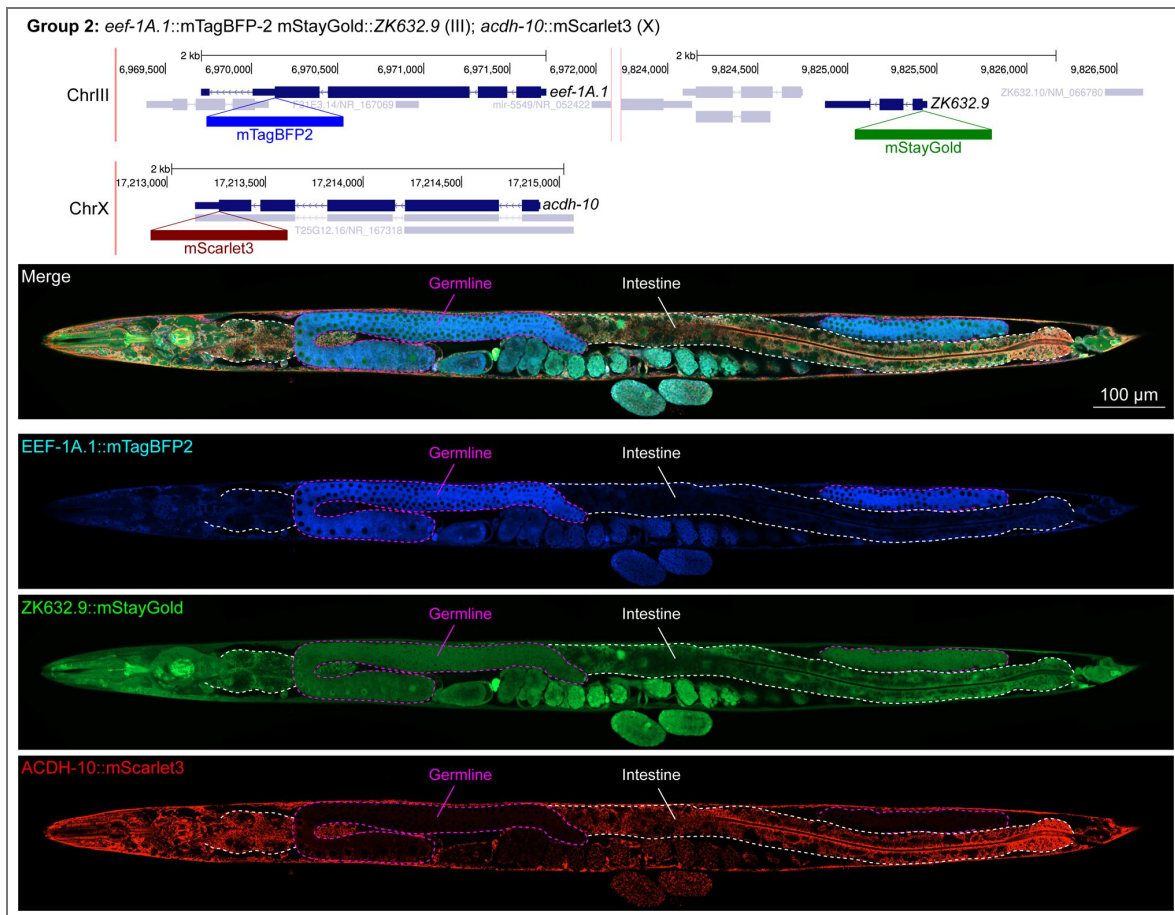


Figure 4. Expression and localization of genes in pool 2.

Expressions and localizations of translation elongation factor 1 alpha 1 EEF-1A.1::mTagBFP2, nuclear protein 1 homolog ZK632.9::mStayGold and acyl-CoA dehydrogenase ACDH-10::mScarlet3 proteins in a live adult hermaphrodite worm by confocal microscopy. All three tags were isolated within the same individual then homozygosed at the F2 generation before imaging. Germline and intestine of the animal are delineated in magenta and white dashed lines, respectively. The worm was imaged in separate overlapping frames with high magnification then stitched to reconstruct the whole animal. A single plane near the mid-plane of the animal was imaged.

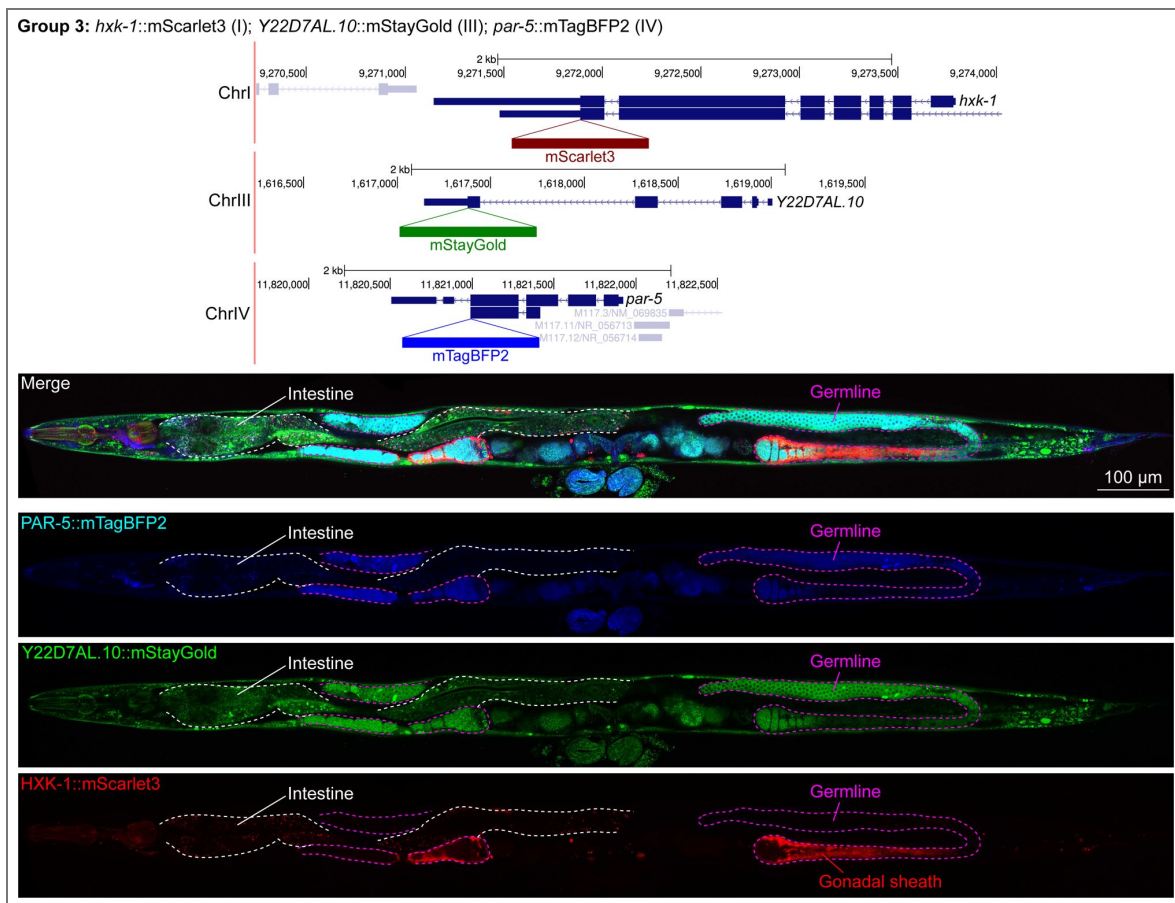


Figure 5. Expression and localization of genes in pool 3.

Expressions and localizations of 14-3-3 protein *PAR-5::mTagBFP2*, heat shock protein family E *Y22D7AL.10::mStayGold* and the glucokinase *HXX-1::mScarlet3* proteins in a live adult hermaphrodite worm by confocal microscopy. All three tags were isolated within the same individual then homozygosed at the F2 generation before imaging. Magenta and white dashed lines, germline and intestine respectively. Image reconstructed from multiple overlapping high magnification images. Single plane near the mid-plane of the animal.

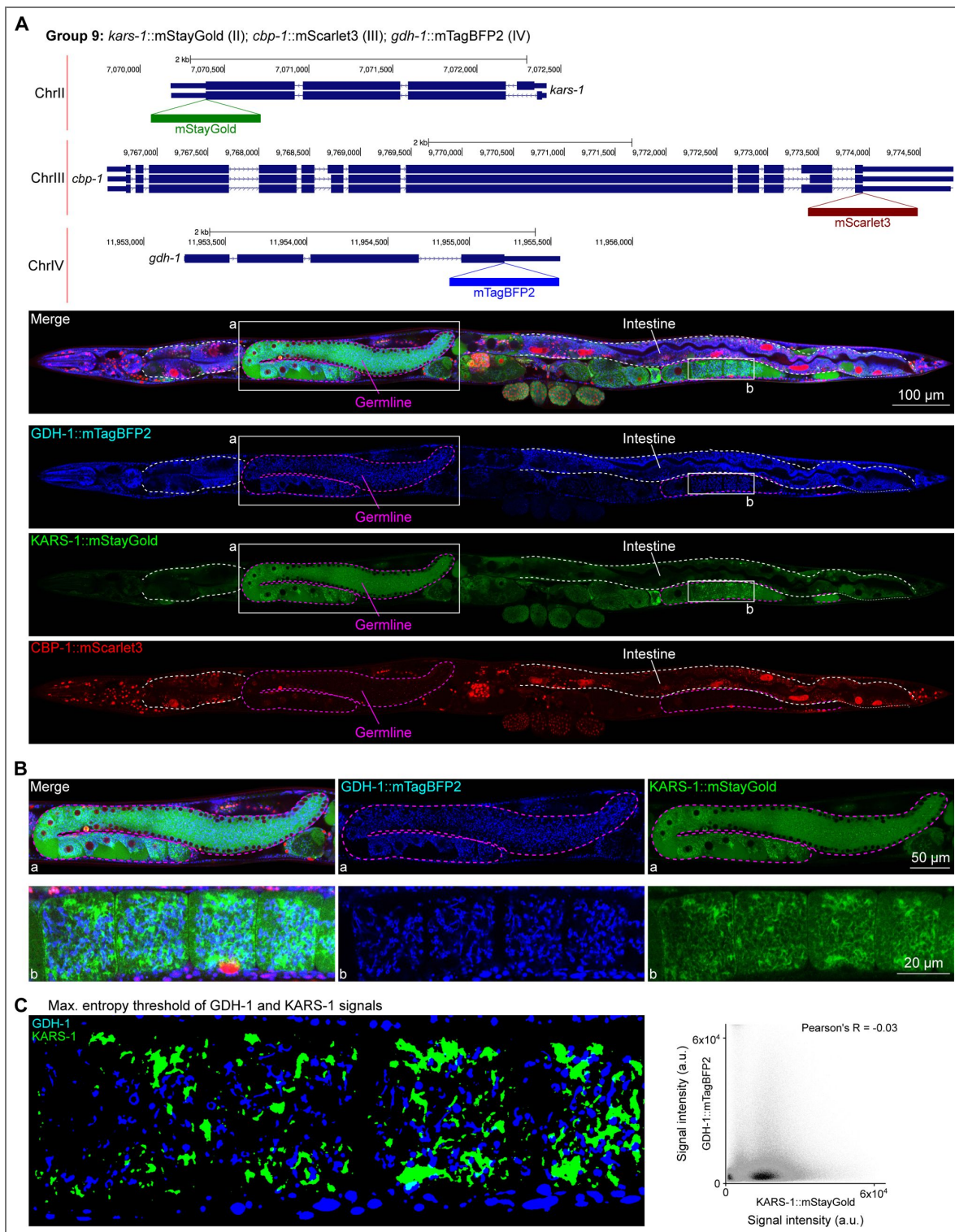


Figure 6. Expression and localization of genes in pool 9.

(A) Expressions and localizations of glutamate dehydrogenase GDH-1::mTagBFP2, lysyl-tRNA synthetase KARS-1::mStayGold and the CREB binding protein CBP-1::mScarlet3 proteins in a live adult hermaphrodite worm by confocal microscopy. All three tags were isolated within the same individual then homozygosed at the F2 generation before imaging. Magenta and white dashed lines, germline and intestine respectively. Image reconstructed from multiple overlapping high magnification images. Single plane near the mid-plane of the animal. **(B)** Germline insets showing non-overlapping localizations of GDH-1 and KARS-1, predicted mitochondrial proteins. **(C)** Left, maximum entropy thresholds of GDH-1 and KARS-1 signals. Right, correlation plot of signal intensities of GDH-1 and KARS-1.

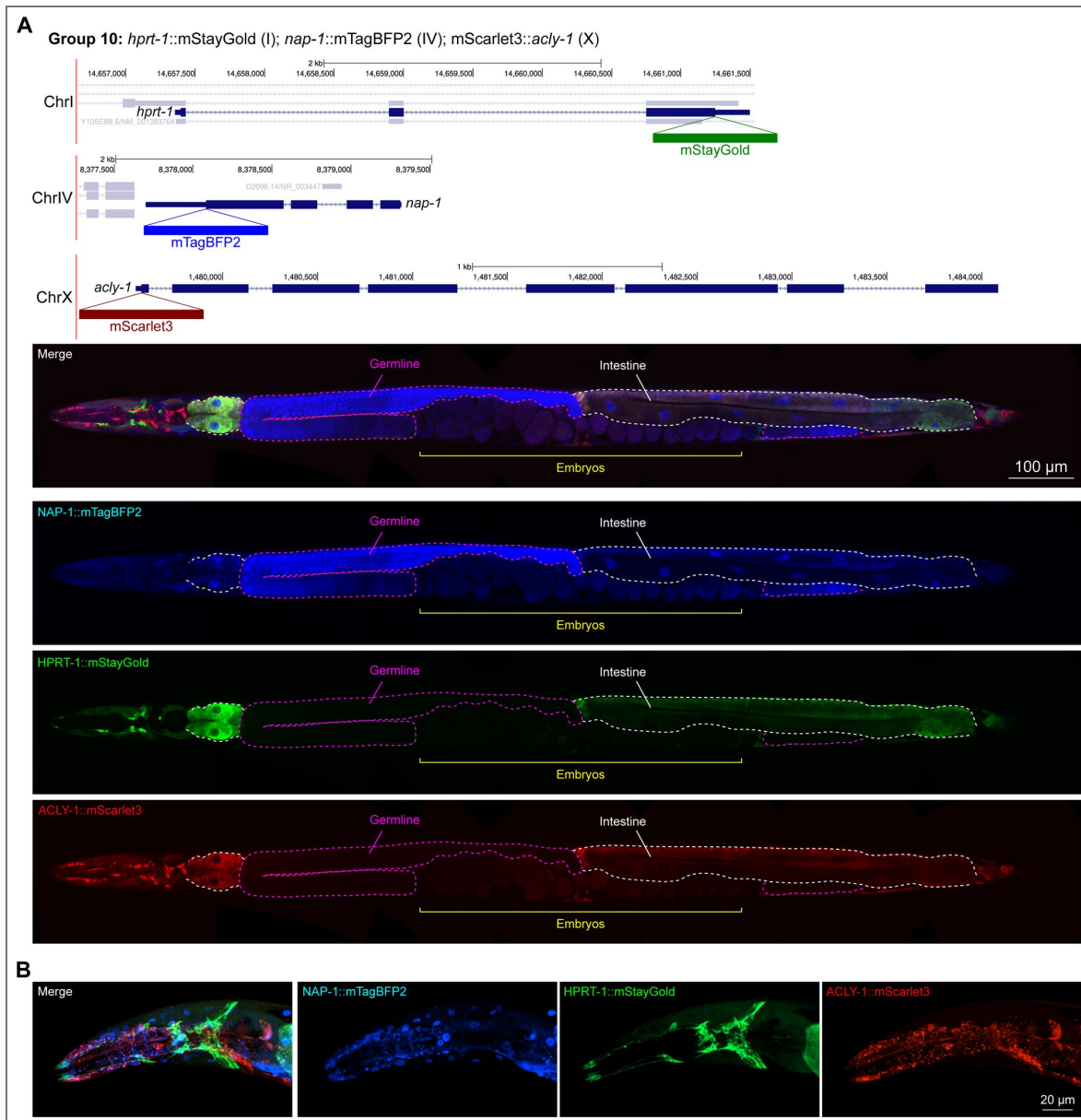


Figure 7. Expression and localization of genes in pool 10.

(A) Expressions and localizations of nucleosome assembly protein *NAP-1::mTagBFP2*, phosphoribosyl transferase *HPRT-1::mStayGold*, and ATP citrate lyase *ACLY-1::mScarlet3* proteins in a live adult hermaphrodite worm by confocal microscopy. Magenta and white dashed lines, germline and intestine respectively. Image reconstructed from multiple overlapping high magnification images. Single plane near the mid-plane of the animal. **(B)** Expressions and localizations of nucleosome assembly protein *NAP-1::mTagBFP2*, phosphoribosyl transferase *HPRT-1::mStayGold*, and ATP citrate lyase *ACLY-1::mScarlet3* proteins in the head of a live adult hermaphrodite worm by confocal microscopy. Image shows a maximum intensity projection of a Z-stack optically sectioned at 1 μm per stack.

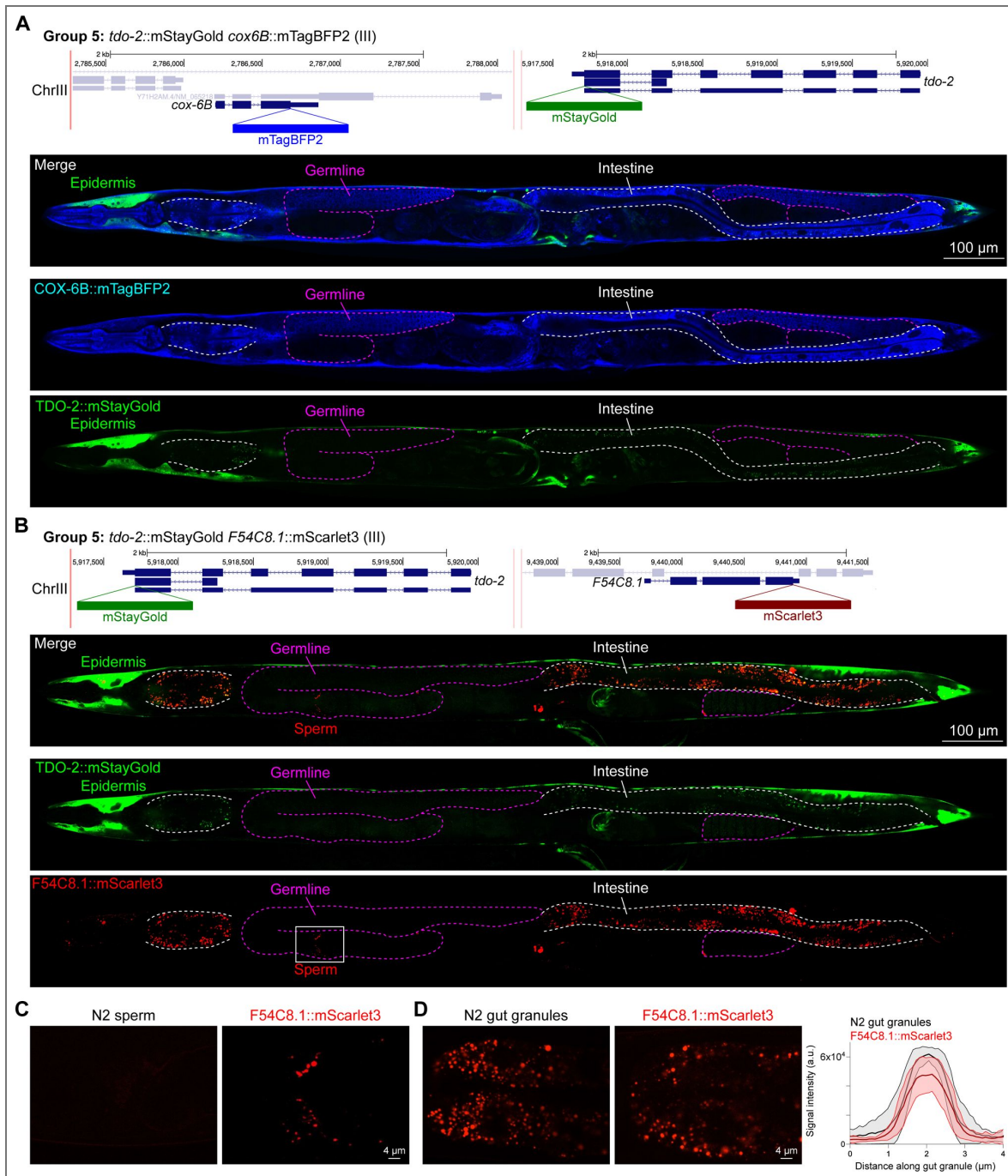


Figure 8. Expression and localization of genes in pool 5.

(A) Expressions and localizations of cytochrome C oxidase subunit 6B COX-6B::mTagBFP2 and TDO-2::mStayGold proteins in a live adult hermaphrodite worm by confocal microscopy. The two tags were isolated within the same individual then homozygosed at the F2 generation before imaging. Magenta and white dashed lines, germline and intestine respectively. Image reconstructed from multiple overlapping high magnification images. Single plane near the mid-plane of the animal. (B) Expressions and localizations of tryptophan 2,3-dioxygenase TDO-2::mStayGold and hydroxyacyl-CoA dehydrogenase F54C8.1::mScarlet3 proteins in a live adult hermaphrodite worm by confocal microscopy. The two tags were isolated within the same individual then homozygosed at the F2 generation before imaging. Magenta and white dashed lines, germline and intestine respectively. Image reconstructed from multiple overlapping high magnification images. Single plane near the mid-plane of the animal. (C) F54C8.1::mScarlet3 signal in sperm compared to untagged N2 worms imaged under the same settings. (D) F54C8.1::mScarlet3 signal in gut granules compared to untagged N2 worms imaged under the same settings. N = 20 F54C8.1::mScarlet3 granules and 24 N2 granules quantified from 2 worms for each genotype. Lines, mean $\pm$ SD.

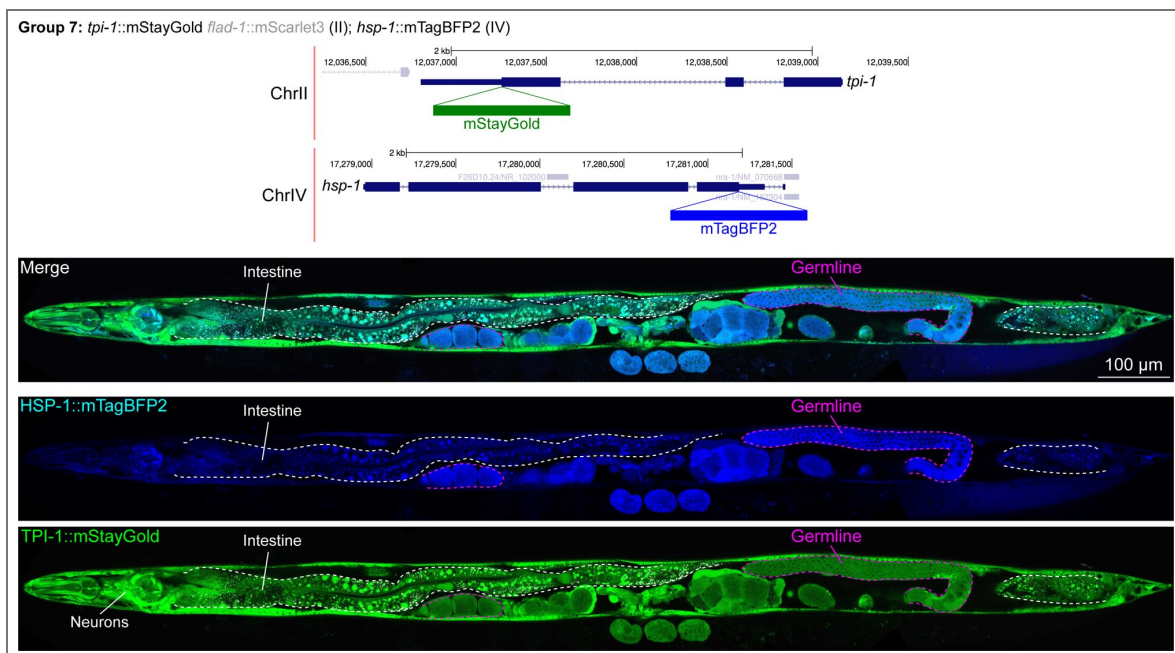


Figure 9. Expression and localization of genes in pool 7.

Expressions and localizations of heat shock protein family A HSP-1::mTagBFP2 and triosephosphate isomerase TPI-1::mStayGold proteins in a live adult hermaphrodite worm by confocal microscopy. The two tags were isolated within the same individual then homozygosed at the F2 generation before imaging. Magenta and white dashed lines, germline and intestine respectively. Image reconstructed from multiple overlapping high magnification images. Single plane near the mid-plane of the animal.

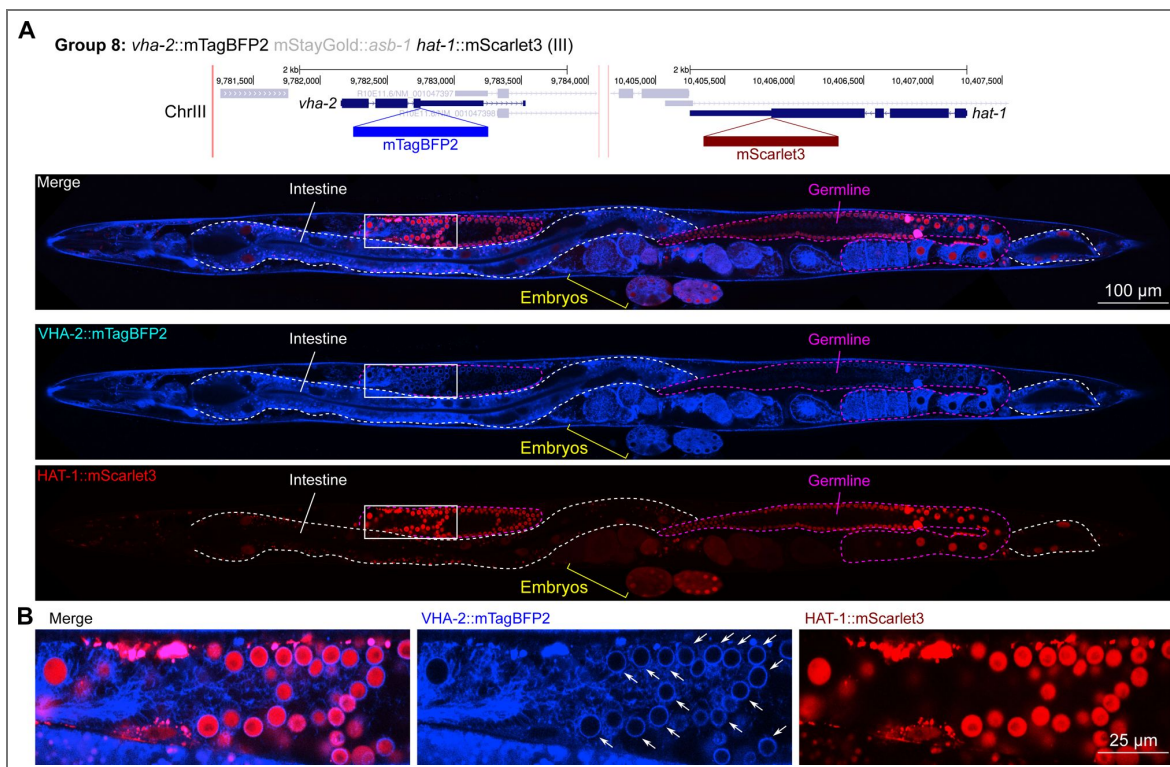


Figure 10. Expression and localization of genes in pool 8.

(A) Expressions and localizations of ATPase H⁺ transporting V0 subunit C VHA-2::mTagBFP2 and histone acetyltransferase HAT-1::mScarlet3 proteins in a live adult hermaphrodite worm by confocal microscopy. The two tags were isolated within the same individual then homozygosed at the F2 generation before imaging. Magenta and white dashed lines, germline and intestine respectively. Image reconstructed from multiple overlapping high magnification images. Single plane near the mid-plane of the animal. **(B)** Inset showing germ cell nuclei. The proton pump VHA-2 localizes to nuclear membranes in germ cells specifically, indicated by white arrows.

viability of tagged worms. We interpret failed tagging attempts as disruptions to protein function, which would necessarily prevent isolation of viable worms, which occurred in 20% of attempted tags. In our experience, inclusion of an 18 amino acid linker reduces the chances of disrupting protein function, and for all our tags we have included a universal 18 amino acid linker. However, we do fully recognize that N- or C-terminal tagging will not always produce properly localized and/or functional protein. We have shown a few examples of functional disruptions here, and it is well appreciated in the field that many types of proteins are difficult to tag without either disruption of function or localization, or both. We also recognize that in some cases where worms appear superficially wild type, tags may have more subtle effects on protein function which may impair fitness. One strategy to predict whether a fused protein may lose function is by comparing the predicted structure of the fusion to the original protein of interest with AlphaFold (Jumper et al., 2021 [↗](#)). However, even if the fused protein is predicted to fold normally, steric interactions, multimeric complex formation and other intrinsic properties of the protein such as solubility may still be affected and are difficult to predict based solely on structure. Smaller tags, including epitope tags like 3xFLAG, V5, and SPOT, can be expected to minimize the risk of such disruptions. However, these present their own problems. Classic epitope tags require fixation of samples, thereby preventing the ability for live imaging and introducing caveats of tissue deformation during fixation and permeabilization. The Halotag system (Los et al., 2008 [↗](#)) has the advantage of also being small and allowing the conjugation of fluorescent dyes, but the need for even dye permeability across cell types and tissues introduces an important limitation. The Alfa tag system that operates through stabilizing fluorescent protein-fused nanobodies (Götzke et al., 2019 [↗](#)) may, for steric reasons, carry the same risk of disruption of endogenous protein function or subcellular localization as covalent tagging. For genes with no obvious mutant phenotypes, assessing the functionality of tagged alleles will be difficult. For such genes, if an interaction-based phenotype such as synthetic lethality is annotated, the tagged gene can be crossed to the interactor to assess synthetic phenotypes. We envision that any novel or unexpected subcellular localizations observed will require validation by independent methods, as in every high-throughput experiment. Overall, the balance between throughput and functionality is, in our view, best struck with translational fusions of fluorophores to proteins of interest, which allows direct live visualization while carrying well understood limitations and artefacts.

Proteins expressed exclusively in a particular cell type and restricted to very specific subcellular sites, such as a handful of synapses in the nerve ring, pose significant challenges for accurate cell identification. Likewise, the cellular origin of secreted proteins may be difficult to trace based solely on translational reporter fusions. In such cases, the labeling of the entire cytoplasm of cells (or, alternatively, its membrane) may be useful. In theory this could relatively easily be achieved by using a double fluorophore tag, where the second fluorophore is separated by a splice leader or T2A sequence from the protein-tagging, first fluorophore. However, in practical terms, such a cassette is much larger than 1 kilobases, and in our experience larger inserts lead to a significant drop in the efficiency of successful insertion, therefore negatively affecting throughput. Where such cases are observed, a small, self-cleaving T2A peptide sequence, which frees up the tag from the fused protein, can be easily inserted in a second round of CRISPR/Cas9 to validate the cell of origin.

Fluorescent landmarks can facilitate delineation of cell boundaries, nuclei, cytoskeleton, or other organelles like mitochondria. Use of visible fluorophores such as mTagBFP2, mStayGold and mScarlet3 leaves the far-red spectrum open for the inclusion of markers such as membrane-targeted far-red fluorescent proteins like mRFP680 (Zhang et al., 2023 [↗](#)) or cell-permeant near-infrared dyes. In an ideal case, pools of target-specific mTagBFP2, mStayGold and mScarlet3 would be directly injected into integrated marker strains expressing mRFP680 targeted to landmarks of interest. Another option is to group genes strategically to include one ubiquitously expressed predicted cytoplasmic gene. With confocal or super-resolution microscopy, such impromptu landmarks may provide some indication of subcellular localization for particularly difficult proteins as a starting point.

Illustrating the discovery aspects of protein tagging, we found that two predicted mitochondrial proteins, KARS-1 and GDH-1, localized to distinct subsets of mitochondria. This is consistent with parallel reports of functionally differentiated mitochondria in cells with varying metabolic demands (Ryu et al., 2024 [DOI](#)). To increase the chance of such discoveries, it may be informative to group multiple proteins predicted to localize to the same subcellular compartments together. With sufficient coverage, this can yield information on the differentiation or uniformity of organelles within the same cell.

We routinely inject 30 animals per day, with 5 worms per injection mix. In theory, this can enable injection of 6 pools targeting 18 genes per day, or 90 per week. A single lab can thus expect 1x coverage of the genome in approximately 290 weeks or 5.6 years, working at a modest pace, assuming that 30% of injections will need to be repeated. With simple quality control steps and careful template preparation, such as ensuring that purity of repair templates is sufficient, we project that fewer injections will need to be repeated. Recent advances in automated injection systems (Pan et al., 2024 [DOI](#)) combined with fluorescent worm sorters may further accelerate coverage. The market cost of ordering crRNAs and annealing with tracrRNA is currently \$76 USD; the cost of each target-specific primer pair for this study was \$23 USD. Approximately \$8 USD of commercially purified Cas9 is used per injection mix; \$1.53 USD for spin columns. Without the one-time small cost of universal fluorophore template synthesis, the material cost of tagging each gene is \$109 USD and would be \$2.2 million USD for the entire genome. While the time, material and personnel costs would be daunting for an individual lab, it would be feasible for a group of committed labs within a few years. A centrally coordinated, but distributed effort would avoid wasteful duplications in gene tagging efforts that are already documented in the literature, such as PGL-1, DAF-16 and several Argonaute genes which each have been tagged at least 4 independent times with GFP (Leyhr et al., 2025 [DOI](#)). The available Caenorhabditis Genetics Center (CGC) strain repository funded by NIH Office of Research Infrastructure Programs (P40 OD010440) as well as emerging database infrastructure dedicated to annotating endogenously tagged strains (Leyhr et al., 2025 [DOI](#)) will facilitate strain as well as reagent (e.g., guide RNA sequence) cataloging and dissemination.

Materials and methods

Nematode maintenance

All *C. elegans* strains were maintained at 20 °C on nematode growth medium (NGM) agar plates with OP50 *Escherichia coli* as a food source (Brenner, 1974 [DOI](#)). All CRISPR/Cas9 insertion strains were generated in the Bristol N2 background.

CRISPR/Cas9-mediated genomic insertions

Sequences of oligonucleotides, fluorophore templates and CRISPR RNAs (crRNAs) are provided in [Supplementary Table 1](#) [DOI](#). Cas9 protein was ordered from Integrated DNA Technologies (Park Coralville, IA, USA) (Alt-R® S.p. Cas9 Nuclease V3, 500 µg, Cat: 1081059), aliquoted at 5 mg/mL concentration, and stored at -80°C until use. tracrRNA was ordered from IDT (Alt-R® CRISPR-Cas9 tracrRNA, 20 nmol, Cat: 1072532), resuspended in the provided nuclease-free duplex buffer to 0.4 µg/µl (18 µM), aliquoted, and stored at -80°C until use (Ghanta and Mello, 2020 [DOI](#)).

All guide RNAs were designed using IDT's guide RNA design tool. The highest 'on-target' scoring guide within 10 bp of the desired insert site was selected and ordered at 2 nmol scale as a custom Alt-R™ crRNA from IDT, resuspended in 20 µL nuclease-free duplex buffer (provided by IDT with tracrRNA order), and stored at 100 µM at -20°C.

Single-stranded DNA repair templates were generated as described (Eroglu et al., 2023 [DOI](#)) with minor modifications. Double-stranded linear DNA templates encoding *C. elegans* codon-optimized mTagBFP2, mStayGold(J), and mScarlet3 were ordered as 'FragmentGENEs' from GENEWIZ (South Plainfield, NJ, USA) and used as universal templates. Primers encoding 45 bp gene-specific homology followed by universal priming sequences (Eurofins Genomics, 10 nmol scale) were used to introduce homology arms to universal fluorophore templates. One primer was phosphorylated

at the 5' end to enable strand-specific digestion by lambda exonuclease. Templates were amplified using Q5 polymerase from New England Biolabs (NEB; M0491) with 0.5 ng of template per reaction at an annealing temperature of 70°C and extension time of 30 seconds with 40 cycles in total. For each gene, a single 50 µL reaction was run. After amplification, individual 50 µL reactions were pooled in sets of three as indicated in Table 1 before proceeding to column purification (PureLink™ PCR Purification Kit, Invitrogen, Cat: K310001). Purified pools of double stranded DNA were digested with lambda exonuclease (NEB, cat: M0262) for 20-30 minutes at 37°C followed by micro-column purification (Monarch® Spin PCR & DNA Cleanup Kit (5 µg)), NEB, Cat: T1130) using a modified protocol described previously (Eroglu et al., 2023). For re-runs of pools 4, 6 and 10 which failed initially, we regenerated the repair templates and ensured that after each column purification, the A260/230 ratio of the purified DNA was ≥ 2.2 and A280/260 was 1.8 ± 0.05 when measured with a Nanodrop spectrophotometer.

Pooled injection mixes were set up by first annealing the crRNAs with tracrRNA as follows: 3.8 µL of tracrRNA and 0.4 µL each of 3 target-specific crRNAs were mixed, then incubated at 95 °C for 5 min followed by 10 °C for 5 min. The Cas9 RNP complex (Paix et al., 2015) was formed by incubating 2.55 µL of tracr-crRNA annealed guide pools with 0.5 µL of Cas9 for 5 min at room temperature. Following Cas9 RNP formation, 4 µL of pooled single stranded DNA template (≥ 400 ng/µL) was added. Constituted injection mixes were injected into *C. elegans* gonads using standard practice. Progeny of injected worms were screened for fluorescence under a Leica M205 FCA fluorescence stereomicroscope with a Planapo 1.6x M-series objective, pE 300 white MB light source and the following filters: Filter set ET DAPI BP - M205FA/M165FC for mTagBFP2; Filter set ET YFP - M205FA/M165FC for mStayGold; and Filter set ET mCherry - M205FA/M165FC for mScarlet3. Worms positive for fluorescence were isolated on individual plates to establish strains.

Confocal microscopy

Worms at the adult stage were immobilized with 50mM sodium azide on 5% agarose pads and imaged on a Zeiss LSM980 laser scanning confocal microscope (C-Apochromat 40x/1.20 water objective) with the 405, 488 and 561nm lasers at variable power settings. Detection was on spectral channels of the 32-channel GaAsP-PMT detector with gain at 800 V for all strains except for group 2 (Fig. 4) which was at 725V. For mTagBFP2, detection wavelengths were set to 411-481 nm; for mStayGold, detection was at 490-552 nm; for mScarlet3, detection was at 570-694 nm.

Multitracking with the LSM980 acousto-optic tunable filter (AOTF) was used to excite and detect mTagBFP2 and mScarlet3 simultaneously on the first light track (405 nm and 561 nm) while mStayGold was excited and detected on a second track (488 nm) with switching between tracks line by line. Scanning was bidirectional with 8x line averaging; pinhole was set to 31 µM (1.00 AU for mScarlet3). Worms were imaged at 1x sampling (213.13 µm x 213.13 µm at 2576 x 2576 pixels) then stitched together on ImageJ (FIJI) (Preibisch et al., 2009; Schindelin et al., 2012) to reconstitute the whole worm. Worms were linearized using the straighten function on ImageJ.

Software and statistical analysis

Genome plots showing gene loci were adapted from the University of California Santa Cruz (UCSC) genome browser (<http://genome.ucsc.edu>) (Casper et al., 2026) and represent ce11 version of the *Caenorhabditis elegans* genome. Plots were generated and statistical analyses performed on GraphPad Prism version 10.6.1.

Data availability

Reported strains are available from the authors upon request. Source data for Figs. 3 and 8D are provided in Supplementary Table 2.

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Additional files

[Supplementary Table 1](#) 

[Supplementary Table 2](#) 

Additional information

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Peer reviews

Reviewer #1 (Public review):

Summary:

Eroglu and Hobert demonstrate that injecting CRISPR guides and repair constructs to target three genes at a time, tagging each with a different fluorescent protein, and selecting which gene to tag with which fluorophore based on genes' expression levels, can improve the efficiency of gene tagging.

Strengths:

This manuscript demonstrates that three genes can be targeted efficiently with three different fluorophores. It also presents some practical considerations, like using the fluorophore least complicated by agar/worm autofluorescence for genes with low expression levels, and cost calculations if the same methods were used on all genes.

Weaknesses:

Eroglu has demonstrated in a previous publication that single-stranded DNA injection can increase the efficiency of CRISPR in *C. elegans* while inserting two fluorescent proteins and a co-CRISPR marker into three loci. The current work is, therefore, an incremental advance. In general, I applaud the authors' willingness to think ahead to how whole proteome tagging

might be accomplished, but I predict that the advance here will be one of many small advances that will get the field to that goal. The title vastly oversells the advance in my view, and the first sentence of the Discussion seems a more apt summary of the key advance here.

Some injections target genes on the same chromosome together, which will create unnecessary issues when doing necessary backcrossing, especially if the mutation rate is increased by CRISPR. Also, the need for backcrossing and perhaps sequencing made me wonder if injecting 3 together really is helpful vs targeting each gene separately, since only 5 worms need to be injected.

The limited utility of current blue fluorescent proteins makes me wonder if it's worth using at all at this stage, before there are better blue (or far red) fluorescent proteins.

Some literature reviews, particularly in the Introduction and Abstract, rely too much on recent examples from the authors' laboratory instead of presenting the state of the field. I'd like to have known what exactly has been done with simultaneous injection targeting multiple loci more thoroughly, comparing what has been accomplished to date by various laboratories' advances to date.

<https://doi.org/10.7554/eLife.110717.1.sa3>

Reviewer #2 (Public review):

The manuscript by Eroglu and Hobert presents a set of strains each harboring up to three fluorescently tagged endogenous proteins. While there is technically nothing wrong with the method and the images are beautiful, we struggled to appreciate the advance of this work - who is this paper for?

As a technical method, the advance is minimal since the first author had already demonstrated that three mutations (fluorophore insertion and co-CRISPR marker) could be introduced simultaneously.

As a pilot for creating genome-scale resources, it is not clear whether three different fluorophores in one animal, while elegantly designed and implemented, will be desired by the broader community.

Finally, the interpretation of the patterns observed in the created lines is somewhat lacking. A Table with all the observations must be included. This can replace the descriptions of the observations with the different lines, which could be somewhat laborious for the reader, and are often wrong. There are numerous mistaken expectations of protein expression here, but two examples include:

(1) The expectation that ACDH-10 is enriched in the intestine and epidermal tissues (hypodermis)

There are multiple paralogs of this protein (see WormPaths or WormFlux) that may share functions in different tissues. There is also no reason to assume that fatty acid metabolism does not occur in other tissues (including the germline). Finally, there are no published studies about this enzyme, so we really don't know for sure what it's doing.

(2) The expectation that HXK-1 is ubiquitously expressed

Three paralogous enzymes are all associated with the same reaction, and we have shown that these three function redundantly in vivo, perhaps in different tissues (PMID: 40011787). Moreover, single-cell RNA-seq data (PMID: 38816550) also show enrichment of *hxx-1* in gonadal sheath cells.

The table should have at least the following information: gene/protein name - Wormbase ID - TPM levels of single cell data assigned to tissues for L2, L4, and adult (all published) - tissues in which expression is observed in the lines presented by the authors.

<https://doi.org/10.7554/eLife.110717.1.sa2>

Reviewer #3 (Public review):

Summary:

The authors argue that establishing the expression pattern and subcellular localisation of an animal's proteome will highlight many hypotheses for further study. To make this point and show feasibility, they developed a pipeline to knock in DNA encoding fluorescent tags into *C. elegans* genes.

Strengths:

The authors effectively make the points above. For example, they provide evidence of two populations of mitochondria in the *C. elegans* germline that differ qualitatively in the proteins they express. They also provide convincing evidence that labelling the whole proteome is an achievable goal with relatively limited resources and time.

Weaknesses:

Cell biology in *C. elegans* is challenging because of the small size of many of its cells, notably neurons. This can make establishing the sub-cellular localisation of a fluorescently tagged protein, or co-localizing it with another protein, tricky. The authors point out in their introduction that advances in light microscopy, such as diSPIM, STED, and ISM (a close relative of SIM), have increased the resolution of light microscopy. They also point out that recent advances in expansion microscopy can similarly help overcome the resolution limit.

(1) Have the authors investigated if the three fluorescent tags they use are appropriate for super-resolution microscopy of *C. elegans*, e.g., STED or SIM? Would Elektra be better than mTAGBFP2? How does mScarlet3-S2 compare to mScarlet 3?

(2) Have the authors investigated what tags could be used in expansion microscopy - that is, which retain antigenicity or even fluorescence after the protocol is applied? It may be useful to add different epitope tags to the knock-in cassettes for this purpose.

The paper is fine as it stands. The experiments above could add value to it and future-proof it, but are not essential. If the experiments are not attempted, the authors could refer to the points above in the discussion.

<https://doi.org/10.7554/eLife.110717.1.sa1>

Author Response:

eLife Assessment

The nematode C. elegans is an ideal model in which to achieve the ambitious goal of a genome-wide atlas of protein expression and localization. In this paper, the authors explore the utility of a new and efficient method for labeling proteins with fluorescent tags, evaluating its potential to be the basis for a larger, genome-wide effort that is likely to be very useful for the community. While the evidence for the method itself is solid, carrying out this project at a large scale will require significant additional feasibility studies.

We appreciate the editor's recognition that the evidence for our method is solid and that a genome-wide protein atlas in *C. elegans* would be highly valuable to the community. However, we respectfully disagree that significant additional feasibility studies are required. As comparison, the yeast proteome-wide GFP tagging project (Huh et al., Nature 2003) achieved ~75% coverage of ~6,000 proteins directly from an established protocol without any prior significant feasibility studies, at least to our knowledge. While the *C. elegans* genome is 3 times in size, we would argue that our tagging protocol may even be less labor intensive as it does not involve any cloning and the screening is visual, requiring no molecular biology skills. Reviewer 3 notes: "They also provide convincing evidence that labelling the whole proteome is an achievable goal with relatively limited resources and time."

Our pilot study validates all key parameters for genome-wide scaling: editing efficiency at novel loci with untested reagents, viability of tagged worms, and detectability of multiple spectrally separated fluorophores across expression ranges. These address the core technical, biological, and practical challenges of large-scale endogenous tagging in a multicellular organism, leaving no fundamental barriers in our view.

The proposed cost and timeline align quite favorably with established large-scale consortium projects: e.g., ENCODE pilot analyzed 1% of the human genome at ~\$55 million over 4 years; Mouse Knockout Consortium scaled to ~20,000 genes over 20 years (ongoing) with ~\$100 million; Human Protein Atlas mapped ~87% of proteins with antibodies in fixed cells (through much more labor intensive methods) over 20+ years at >\$100 million. With ~8% of *C. elegans* genes already tagged (WormTagDB), scaling our protocol to the proteome is feasible, potentially covering the genome in 5-6 years by a single lab or faster with distributed effort at a reagent cost of merely \$2.2 million. The main barriers now are funding commitment and assembling collaborators, not further feasibility testing.

Public Reviews:

Reviewer #1 (Public review):

Summary:

Eroglu and Hobert demonstrate that injecting CRISPR guides and repair constructs to target three genes at a time, tagging each with a different fluorescent protein, and selecting which gene to tag with which fluorophore based on genes' expression levels, can improve the efficiency of gene tagging.

Strengths:

This manuscript demonstrates that three genes can be targeted efficiently with three different fluorophores. It also presents some practical considerations, like using the fluorophore least complicated by agar/worm autofluorescence for genes with low expression levels, and cost calculations if the same methods were used on all genes.

Weaknesses:

*Eroglu has demonstrated in a previous publication that single-stranded DNA injection can increase the efficiency of CRISPR in *C. elegans* while inserting two fluorescent proteins and a co-CRISPR marker into three loci. The current work is, therefore, an incremental advance. In general, I applaud the authors' willingness to think ahead to how whole proteome tagging might be accomplished, but I predict that the advance here will be one of many small advances that will get the field to that goal.*

Our manuscript indeed builds on prior multiplex editing (including our own co-CRISPR work), but the manuscript's primary contribution is not a novel technical breakthrough *per se*. Instead, our main goal was to pilot and strategize a feasible path to whole-proteome

tagging in *C. elegans* and importantly test the following key parameters: (1) success rate of triple pools with prior untested reagents at novel targets; (2) utility of fluorophores across expression levels; (3) major effects on tagged protein function. In prior multiplexing, we used two targets which we already knew could be edited quite efficiently, with the 3rd target a point mutation with nearly 100% efficiency. Thus, it was not at all clear that picking 3 random genes and replacing the 3rd highly efficient locus with another less efficient large insertion would work or be sufficiently scalable for thousands of novel genes with unvalidated reagents at first pass.

The title vastly oversells the advance in my view, and the first sentence of the Discussion seems a more apt summary of the key advance here.

Some injections target genes on the same chromosome together, which will create unnecessary issues when doing necessary backcrossing, especially if the mutation rate is increased by CRISPR.

We disagree with the reviewer's assessment of the need for backcrossing, for two reasons: (1) Prior studies have shown that off-target mutations are not a serious concern in *C. elegans* (reviewed in PMID: 26336798 and PMID: 24685391). For instance, WGS of strains after CRISPR/Cas9 found negligible off-target effects (PMID: 25249454, PMID: 30420468 – using similar RNP/ssDNA method and multiple guides; PMID: 23979577, PMID: 27650892 using other methods). Targeted sequencing studies have reported similar findings, using various CRISPR/Cas9 methods, with essentially no mutations at sites other than the intended target (PMID: 23995389; PMID: 23817069). (2) If the goal is to tag the entire genome, the introduction of backcrossing should not reasonably be a routine part of the initial tagging.

Lastly, if one wants to backcross at a later stage, the existence of tags on the same chromosome is actually an advantage because it permits selection for recombinants with wild-type chromosomes.

Also, the need for backcrossing and perhaps sequencing made me wonder if injecting 3 together really is helpful vs targeting each gene separately, since only 5 worms need to be injected.

Apart from our disagreement regarding backcrossing, we are puzzled by the reviewer's comment that tagging each gene separately may not be considered helpful. Why would one do single tagging at a time, rather than triple tagging if the whole point of the paper is to demonstrate the scalability of tagging? Meaning, that one can shortcut tagging all genes by a factor of 3 through joint tagging? It is important to keep in mind that the rate limiting step for tagging the whole genome is the number of injections that can be done per day. Since there is no cloning to generate the repair templates/guides and all other reagents are commercially available and not sample specific, these can be prepared quite rapidly. Being able to isolate multiple lines (together or independently) from the same injection increases throughput 3-fold and in our view does not provide any disadvantages as individual tags can be isolated independently if desired.

Beyond the numerous technical advantages pooling provides (also lower cost and throughput for making injection mixes as well as imaging), our results show that it yields epistemic benefits as well: we would never have noted the subcellular pattern in Fig. 6B, C with different sets of mitochondria being marked by different mitochondrial proteins had we imaged them separately or even aligned to a pan-mitochondrial landmark. As we mentioned in the discussion, grouping proteins predicted to localize to the same compartment together can simultaneously test how uniform or differentiated such compartments are during the screen.

The limited utility of current blue fluorescent proteins makes me wonder if it's worth using at all at this stage, before there are better blue (or far red) fluorescent proteins.

We do not think that the utility of current BFPs is very limiting. The theoretical brightness of mTagBFP2 is comparable to that of EGFP (PMID: 30886412), which was useful for the bulk of currently tagged proteins. Due to modestly higher autofluorescence in the blue spectrum, the practical brightness is somewhat less ideal, but we have shown that many proteins are expressed high enough to be detected quite well with mTagBFP2 by eye at low magnification. We also note that many tags that are not visible by eye under a dissection scope become visible with long exposure cameras of widefield microscopes or modern confocal (GaAsP) detectors, so the list of genes detectable with mTagBFP2 is likely to be much higher. We routinely use mTagBFP2 to super-resolve subnuclear structures with endogenous tags (e.g., in the nucleolus), with some tags having lower annotated FPKMs than the genes tested here.

Some literature reviews, particularly in the Introduction and Abstract, rely too much on recent examples from the authors' laboratory instead of presenting the state of the field. I'd like to have known what exactly has been done with simultaneous injection targeting multiple loci more thoroughly, comparing what has been accomplished to date by various laboratories' advances to date.

We are not sure what the reviewer is referring to when bemoaning that the Abstract and Introduction are too focused on our paper and not presenting the state of the field. In the Abstract, we do not refer to any literature. In the Introduction, we cite 28 papers, 6 of those from our lab (4 of which providing examples of protein tags). We do not believe that this can be fairly called an unbalanced presentation of the state of the field.

This being said, we will gladly expand our Introduction to provide more background on co-CRISPRing. Labs have routinely used co-conversion (“coCRISPR”) markers for picking out their intended edits (e.g., point mutations or insertions), as it has been shown by multiple groups that a CRISPR/Cas9 edit at one locus correlates with efficiency at other simultaneous targets (PMID: 25161212). Generally, making point mutations with the Cas9/RNP protocol is highly efficient, especially at specific loci such as *dpy-10*. However, multiple FP-sized insertions have not been routinely attempted. We and only one other group have successfully attempted it using previously working targets and reagents (e.g., 28% in PMID: 26187122). Importantly, the efficiency of such multiple insertions has never been assessed at scale and using entirely untested reagents at novel sites – critical parameters to determine for a whole genome approach. So, we test here (1) the efficiency of triple insertions and (2) the chance of getting them with new and untested guides and reagents.

In our view, since we have to use some injection/coCRISPR marker anyway for those genes which are not expressed at dissecting-scope visible levels (likely most genes), using highly expressed intended targets as improvised markers in a pooled approach makes our approach much more efficient. It allows us to find the worms with the highest chance of yielding CRISPR insertions, which we can screen with higher power methods for the dimmer targets, while enabling us to co-isolate other intended targets. Insertions, being often heterozygous in F1, can be segregated independently if desired, or homozygosed together to facilitate maintenance then outcrossed individually by those interested in studying specific genes in more detail.

In the revised version of this manuscript, we will discuss some of these points in the first paragraph of the results section:

“In *C. elegans*, screening for novel CRISPR/Cas9-induced genomic edits is facilitated either by use of co-injection markers (i.e., plasmids that form extrachromosomal arrays) that yield phenotypes or fluorescence in progeny of successfully injected worms, or co-editing well

characterized loci using established and highly efficient reagents which likewise yield visible phenotypes. In the latter approach, termed “co-CRISPR”, worms edited at the marker locus are most likely to also carry the intended edit (Arribere et al., 2014).”

“These attempts pooled reagents previously established to work efficiently and targeted genes that were known to yield functional fusion proteins when tagged. Thus, while in principle current methods could allow tagging of at least 3 independent loci in one injection if a co-CRISPR marker is omitted, it is not known to what extent such an approach could be generalized across the genome with previously unvalidated reagents (i.e., guides and repair template homology arms) at novel loci.”

Reviewer #2 (Public review):

The manuscript by Eroglu and Hobert presents a set of strains each harboring up to three fluorescently tagged endogenous proteins. While there is technically nothing wrong with the method and the images are beautiful, we struggled to appreciate the advance of this work - who is this paper for?

We consider this paper to have two purposes: (1) motivate the community to come together to consider such genome-wide tagging approach; (2) provide a reference point for funding agencies that such an aim is not unreasonable and will provide novel interesting insights.

As a technical method, the advance is minimal since the first author had already demonstrated that three mutations (fluorophore insertion and co-CRISPR marker) could be introduced simultaneously.

We agree that the basic principle is similar. However, it was not clear that triple pooling three novel large edits would work, given the numbers in our original paper or that it would be scalable.

The *dpy-10* coCRISPR marker previously used is a highly efficient single site, with close to 100% hit rate. We also knew in the earlier study that the two pooled insertions already worked quite efficiently and did not disrupt the function of targeted proteins. Exchanging these plus *dpy-10* for three novel tags was not guaranteed to succeed for many potential reasons, including both biological and technical. For instance, such a “marker free” approach necessitates that a significant number of targets in the genome should be expressed highly enough to be visible by fluorescence stereomicroscopy when tagged with current best fluorophores. The chance of disrupting gene function by tagging was also not explored in detail in *C. elegans*, nor whether one untested guide is generally sufficient. We think that establishing these parameters was meaningful and necessary for the goal of whole genome tagging. We have clarified some of these points in the text.

As a pilot for creating genome-scale resources, it is not clear whether three different fluorophores in one animal, while elegantly designed and implemented, will be desired by the broader community.

The usage of three different fluorophores is largely driven by the ability to co-inject and therefore cut injection effort by a factor of three. Moreover, having all three fluorophores together facilitates imaging and maintenance. Lastly, co-labeling has the potential to reveal unexpected patterns of co-localization or lack thereof (example: two mitochondrial proteins that we found to not have overlapping distribution). We clarified this point in the revised text in both the results and discussion.

Finally, the interpretation of the patterns observed in the created lines is somewhat lacking. A Table with all the observations must be included. This can replace the descriptions of the observations with the different lines, which could be somewhat

laborious for the reader, and are often wrong. There are numerous mistaken expectations of protein expression here, but two examples include:

We are not convinced that expectations are mistaken. Below we respond to the reviewer's specific examples and we are open to hear from the reviewer about additional cases.

(1) The expectation that ACDH-10 is enriched in the intestine and epidermal tissues (hypodermis).

There are multiple paralogs of this protein (see WormPaths or WormFlux) that may share functions in different tissues. There is also no reason to assume that fatty acid metabolism does not occur in other tissues (including the germline). Finally, there are no published studies about this enzyme, so we really don't know for sure what it's doing.

The expression of *acdH-10* is annotated in multiple scRNA datasets as intestine and epidermal enriched (Packer et al 2019, highest intestine and hyp; Ghaddar et al 2023 intestine, sheath and BWM, and even oocyte). We did not mean to imply that fatty acid metabolism does not occur in the gonad, nor that a paralog of *acdH-10* could not be performing the same function in tissues where *acdH-10* is not expressed.

However, this raises an important question: why have different paralogs doing the same thing? Duplicate genes with the same function are generally not evolutionarily stable (PMID: 11073452, PMID: 24659815). That there are such striking tissue specific expression patterns of an essential or widely expressed protein class suggests that paralogs of the gene likely differ in some meaningful parameter that might align with tissue-specific functional needs or regulation. The reviewer's statement that "there are no published studies about this enzyme, so we really don't know for sure what it's doing" is in fact an excellent demonstration of our point; finding out where the duplicates are expressed can provide a starting point to uncover potential differences between the paralogs. At the very least it can delineate to what degree paralogs diverge in their expression across the proteome and identify which such cases merit further study. In a more ideal scenario, prior information of protein function could indicate that the involved pathway requires tissue specific regulation.

(2) The expectation that HXK-1 is ubiquitously expressed.

Three paralogous enzymes are all associated with the same reaction, and we have shown that these three function redundantly in vivo, perhaps in different tissues (PMID: 40011787).

The cited paper (PMID: 40011787) does not show where they are expressed. We discussed redundancy/paralogs above in point 1, and in our view the same applies here. They may perform the same reaction but are likely to differ in some meaningful way, be it regulation or rate of activity, for them to be stably maintained as functional genes over evolution.

*Moreover, single-cell RNA-seq data (PMID: 38816550) also show enrichment of *hXK-1* in gonadal sheath cells.*

We note that the Ghaddar et al. and CeNGEN/Taylor et al. datasets do not. The scRNA paper cited by the referee (PMID: 38816550) also shows enrichment in neurons and pharynx, which we did not note. In our view, these in fact further support our goals: often, transcript datasets alone (frequently used to infer tissue function) do not sufficiently predict protein expression. One can *post hoc* find an scRNA-seq dataset that aligns somewhat with our protein observations, but how does one know which to trust *a priori*? Disagreements between transcript datasets will ultimately require resolution at the protein level, in our view.

To clarify these points, we will add the following to the discussion section:

“We also noted unexpected cell type dependent distributions of proteins involved in broadly important metabolic processes such as *ACDH-10*, which was depleted from the germline compared to other tissues, and *HXK-1*, which was highly enriched in the gonadal sheath. Notably, for these as well as other cases, scRNA-seq datasets were not sufficient to deduce *a priori* the observed cell type specific differences at the protein level. Importantly, many genes encoding metabolic enzymes including *acdH-10* and *hXK-1* have paralogs that likely perform similar catalytic functions. Yet, duplicate genes with identical functions are generally not evolutionarily stable (Adler et al., 2014; Lynch and Conery, 2000); thus such genes are likely to differ in some meaningful parameter (e.g., regulation or activity) that might align with tissue-specific functional needs. Fully annotating the expression patterns of paralogs at the protein level could indicate which tissues require unique metabolic needs and indicate which paralogous genes have undergone sub- versus neo-functionalization. For those proteins that are less functionally understood, unexpected distributions might indicate which merit further study.”

The table should have at least the following information: gene/protein name - Wormbase ID - TPM levels of single cell data assigned to tissues for L2, L4, and adult (all published) - tissues in which expression is observed in the lines presented by the authors.

We will add this information to the table including annotated expression levels in young adults from various datasets (but not larval datasets as we did not image these). We note that each of these studies use different pipelines and report different metrics (scaled TPM/Z-score versus Seurat average expression versus TPM), so comparisons between them are not informative unless they are integrated and analyzed together.

Reviewer #3 (Public review):

Summary:

*The authors argue that establishing the expression pattern and subcellular localisation of an animal's proteome will highlight many hypotheses for further study. To make this point and show feasibility, they developed a pipeline to knock in DNA encoding fluorescent tags into *C. elegans* genes.*

Strengths:

*The authors effectively make the points above. For example, they provide evidence of two populations of mitochondria in the *C. elegans* germline that differ qualitatively in the proteins they express. They also provide convincing evidence that labelling the whole proteome is an achievable goal with relatively limited resources and time.*

We are grateful for the referee's appreciation that whole proteome tagging is feasible.

Weaknesses:

*Cell biology in *C. elegans* is challenging because of the small size of many of its cells, notably neurons. This can make establishing the sub-cellular localisation of a fluorescently tagged protein, or co-localizing it with another protein, tricky. The authors point out in their introduction that advances in light microscopy, such as diSPIM, STED, and ISM (a close relative of SIM), have increased the resolution of light microscopy. They also point out that recent advances in expansion microscopy can similarly help overcome the resolution limit.*

*(1) Have the authors investigated if the three fluorescent tags they use are appropriate for super-resolution microscopy of *C. elegans*, e.g., STED or SIM? Would Elektra be better than mTAGBFP2? How does mScarlet3-S2 compare to mScarlet 3?*

All three tags work for ISM (i.e., Airyscan). We previously tried Electra (not for the genes tested here) but could not isolate positive tags. Given Electra is not that much brighter on paper than mTagBFP2 we did not pursue it further, though we recognize that these may simply have been unlucky injections. mScarlet3-S2 is quite a bit dimmer than mScarlet3 on paper – the advantage is that it has higher photostability. In our view, the limiting factor will be having FPs that are bright enough to screen, image and scale to the whole genome, so brightness will likely provide an advantage over photostability at this stage.

(2) Have the authors investigated what tags could be used in expansion microscopy - that is, which retain antigenicity or even fluorescence after the protocol is applied? It may be useful to add different epitope tags to the knock-in cassettes for this purpose.

mSG and mSc3 retain fluorescence after fixing with formaldehyde. We have not tested mTagBFP2 fluorescence in fixed worms. We agree that adding different epitope tags would be useful.

The paper is fine as it stands. The experiments above could add value to it and future-proof it, but are not essential. If the experiments are not attempted, the authors could refer to the points above in the discussion.

<https://doi.org/10.7554/eLife.110717.1.sa0>